

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 18)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

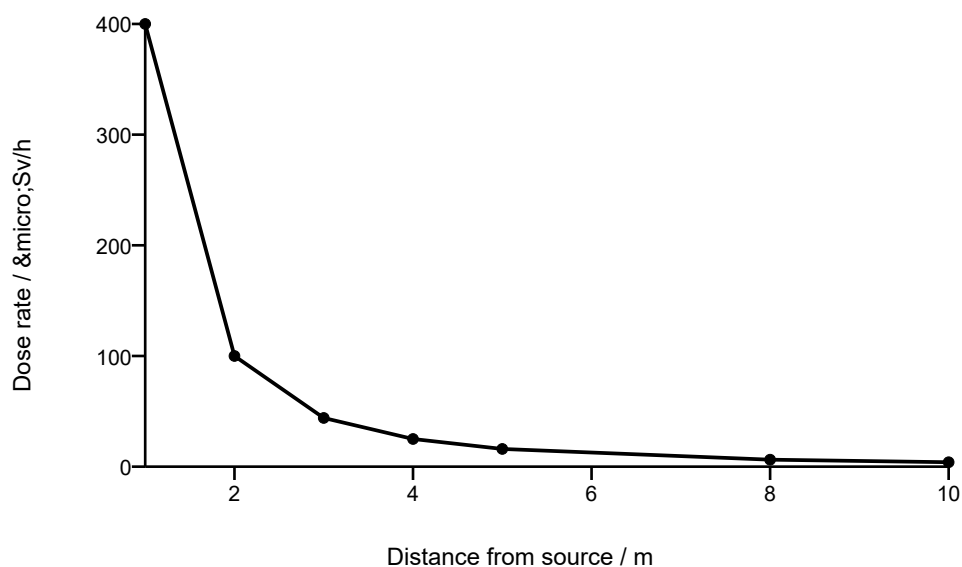
Question	Mark
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows the radiation dose rate measured at different distances from a small gamma source.

Figure 1



0 1

1

Use Figure 1 to show that the dose rate follows the inverse square law.

[2 marks]

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0 1

2

Calculate the dose rate at 20 m from the source.

[1 mark]

Answer

$\mu\text{Sv/h}$

0 1

3

Explain how each of the following reduces the exposure of workers handling radioactive sources.

[4 marks]

Closed (sealed) sources

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Absorbing materials

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Reduced exposure time

.....

Worker monitoring

.....

0 1 . 4 Explain why potassium iodide tablets are issued to people near a nuclear accident.

[2 marks]

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0 1 . 5 State the meaning of ALARA.

[1 mark]

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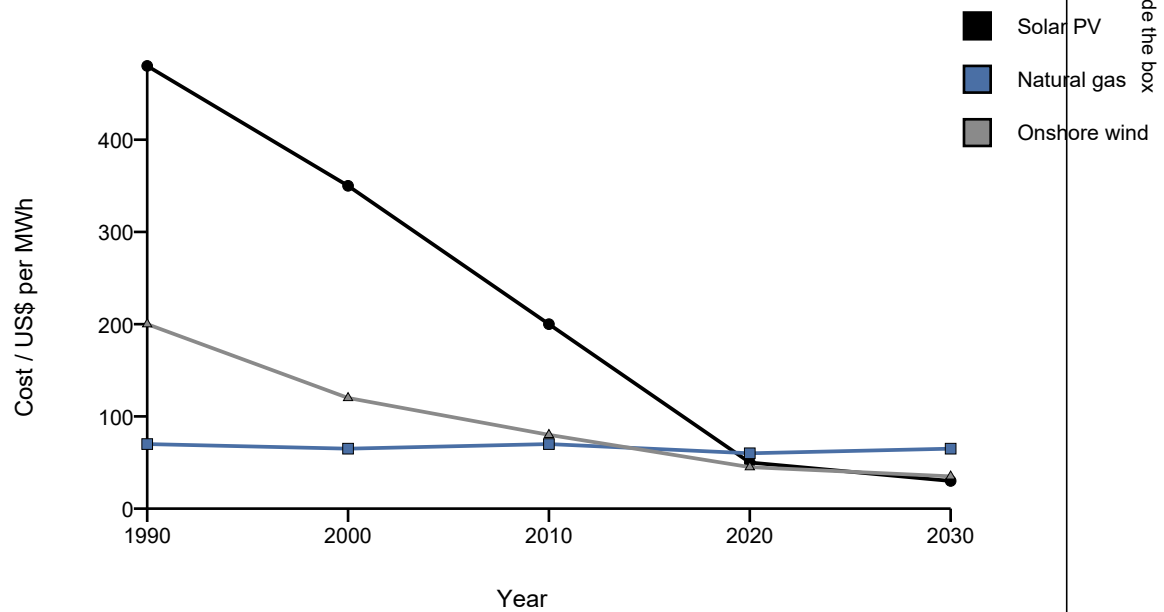
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10

0 2

Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 2

1 Describe the trends shown in Figure 1.

[3 marks]

0 2

2 Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

0 2 . 3

Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

0 2 . 4

Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

10

0 3 . 1

Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation

Urban development

Agriculture

03 . 2

Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

1

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2

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03 . 3

Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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10

04 . 1

Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

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0 4 . 2

Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

0 4 . 3

Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

10

0 5

Table 1 lists some techniques used in ecological monitoring.

0 5 . 1

Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation
Image databases of unique markings		
eDNA sampling		
Geolocator tags		

0 5 . 2

Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

0 5 . 3

Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

1

2

10

0 6 . 1

Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients ☒ to the soil
- B It produces more electricity ☐
- C It removes heavy metals ☐
- D It destroys asbestos fibres ☐

0 6 . 2

Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

06 . 3

Give **two** features of a well-managed landfill site that reduce pollution.**[2 marks]**

1

2

5

07

Table 1 compares the energy used to produce 1 tonne of aluminium from bauxite ore with the energy used to produce 1 tonne from recycled aluminium cans.

Table 1

Source	Energy used / GJ per tonne	CO ₂ emitted / tonnes per tonne of aluminium
Bauxite ore (extraction, refining, smelting)	210	12
Recycled cans (collection, sorting, remelting)	12	0.7

07 . 1

Calculate the percentage energy saving from recycling aluminium rather than extracting it from ore.

[2 marks]

Answer

%

07 . 2

Give **three** advantages of recycling metals other than energy saving.**[3 marks]**

1

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2

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3

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07 . 3 Explain **four** difficulties that reduce the effectiveness of recycling schemes. **[4 marks]**

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07 . 4 Explain what is meant by **Cradle to Cradle** design. **[1 mark]**

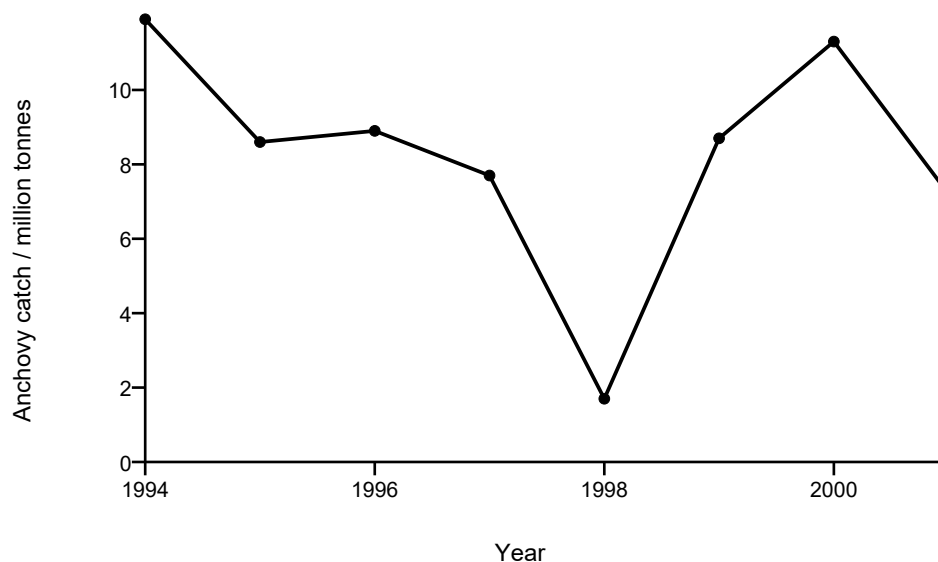
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10

0 8

Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1**0 8****1**

Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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0 8**2**

Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

[3 marks]

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08 . 3

Describe **two** other effects of El Niño events.

[2 marks]

1

2

08 . 4

Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

10

09 . 1

Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

09 . 2

Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

09 . 3

Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

09 . 4

Explain what is meant by **synergism** and give an example.

[3 marks]

10

10 . 1

Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

[illegible]

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10 . **2** State **one** effect of tropospheric ozone on plants.

[1 mark]

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10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2

Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dotted lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

Do not write outside the box

END OF QUESTIONS